DUNMAN HIGH SCHOOL
PRELIMINARY EXAMINATION
Year 6

HIGHER 2 GEOGRAPHY PAPER 2 **9173/02**

INDICATIVE CONTENT

1 (a) With reference to the research question, explain if it is of a suitable scale. [3]

Suitable Scale

- **Spatial scale**: The river is close enough to the school [**1 mark**].

May not be suitable in scale

- Manpower constraints [**1 mark**]
- Resource limitation [**1 mark**]

(b) Explain one strength and one limitation of using each method in measuring river velocity shown in Resources 2A and 2B. [4]

Resource 2A: Float Method

Strength: The float method is cost effective [**1 mark**]

Limitation: The float method can be less accurate than electronic instruments. [**1 mark**]

Resource 2B: Flowmeter

Strength: More accurate than using float method [**1 mark**]

Limitation: Proper placement of the flowmeter can be difficult in challenging [**1 mark**]

(c) Suggest issues related to research ethics and how the students can address them in this geographical investigation. [6]

Issue: This could be a breach of rights [**1 mark**]. **To Address**: Obtain permission [**1 mark**]

Issue: Could harm wildlife [**1 mark**]. **To Address**: Removing any rubbish produced [**1 mark**]

Issue: When students release the float, this can cause pollution [**1 mark**]. **To Address**: use biodegradable floats [**1 mark**]

(d) Suggest the considerations that the students should have in the selection of the investigation sites and the sampling method. [5]

<u>Considerations</u>

There may be sections which are slippery. As such, they should have proper footwear with grip [**1 mark**]. Some sections may be murky posing a drowning risk. Students should be proficient in water safety [**1 additional mark**]

Site Sampling Consideration

Due to the length of the river, the students can divide the river into segments to ensure less biased sampling [**1 mark**]. Students should also consider characteristics such as different land uses [**1 mark**]. Students can also consider variation in natural characteristics [**1 mark**].

(e) With reference to Resource 3A, describe the measurements recorded. [6]

The average velocity for float method are all higher than flowmeter method in all sites [**1 mark**] except for the left channel at Site C where the float method is slower [**1 additional mark**]

The velocity at the centre of the channel tends to be the highest [**1 mark**] except for the float method at Site B [**1 additional mark**]

The average velocity tends to be higher at Site C compared to Site A [**1 mark**] except for the reading derived by the flowmeter method for Site B [**1 additional mark**].

(f) Suggest two limitations of the method of data representation in Resource 3B. [2]

Tables do not effectively convey gradual changes / variations [**1 mark**] Difficult to visually assess differences without a graphical representation [**1 mark**]

(g) Sketch the cross-section at Site C shown in Resource 3B. [4]

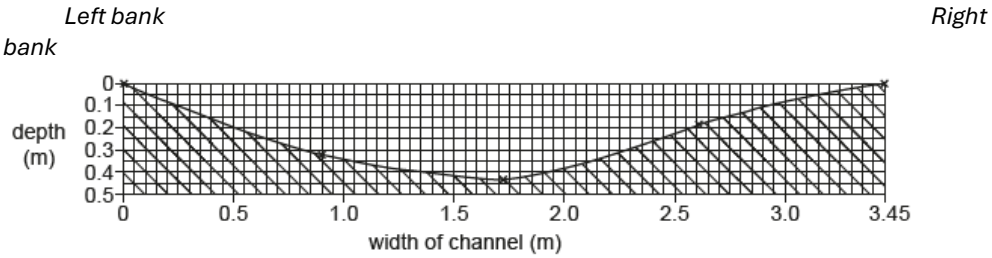


Fig 1: Cross-section at Site C

- 1 mark: Data used and labelling of banks
- 1 mark: labelling of axis (including units)
- 1 mark: plotting / accuracy (connecting the points)
- 1 mark: title

(h) Evaluate the students' investigation of flood risk along the river. [10]

Data Collected (Validity)

Candidates could refer to the Resources and consider how accurate and reliable the collected data are for drawing conclusion. Candidates could look at the way data are collected and assess if the methods used could make accuracy and reliability of the data questionable. For instance, they could look at whether the collection of data over one day at one timing and in only one part of the year is sufficient in providing robust data for their investigation. Information from Resources and preamble could be used to infer the factors that could contribute to making the data in other Resources inaccurate and unreliable.

2 (a) Describe the relationships between the National Ecological footprint, Human Development Index and level of urbanisation shown in Resource 4. [5]

There is a **positive correlation** between HDI, EF and urbanisation level [**1 mark**] with an **exponential pattern** observed between HDI value and EF [**1 additional mark**].

Countries at high levels of urbanisation have a **narrow HDI range** but a **large EF range** [**1 mark**]

Countries at low levels of urbanisation tended to have a **wide HDI range** but a **small EF range** [**1 mark**]. In contrast, countries with low levels of urbanisation tend to have a wide HDI range of and a narrow EF range [**1 additional mark**]

Anomalies
However, there are three countries with high levels of urbanisation (61-70% and above 90%) that has a lower HDI (0.55) [**1 mark**]

(b) Suggest reasons for the relationship between level of urbanisation and national ecological footprint observed in Resource 4. [6]

High level of urbanisation correlated with higher level of EF

Activities in urban areas produce **large amount of waste** [**1 mark**].

Urbanites tend to be more **affluent,** consuming more resources [**1 mark**]. In addition, most of these resources are also sourced from hinterland leading to a higher ecological footprint [**1 additional mark**].

A **lack of legislation** on effective recycling may also lead to higher ecological footprint [**1 mark**].

Countries with high levels of urbanisation may also be correlated with lower ecological due to **stronger legislation** [**1 mark**].

A lower level of urbanisation does not consume as much resources for its economic activities as it may not be as industrialised [**1 mark**].

(c) With reference to Resources 5A and 5B, explain how cities can reduce their ecological footprint and achieve sustainable urban development. [4]

By optimising the use of materials in urban processes through technologies, this can help **reduce the total input** used [**1 mark**]. Instead of viewing waste as a final product, cities can adopt practices like **recycling inorganic and organic waste**, reducing the total input used [**1 mark**].

Through incinerating less or using less landfills, less emissions will be produced, thereby helping to mitigate climate change and achieving environmental sustainability **[1 mark**]. The use of renewable energies can help promote environmental sustainability through the reduction of carbon emission from traditional fossil fuel, mitigating climate change [**1 additional mark**].

(d) With reference to Resource 6, explain the usefulness and limitations of the City Development Index (CDI) in monitoring sustainable urban development. [5]

It provides a fairly comprehensive assessment of sustainable urban development in assessing social sustainability. [**1 mark**]. For instance, sub-indices which measures access to basic utility, education and healthcare [**1 additional mark**].

Knowing the score of the different components allow cities to know the area which requires more effort devoted to achieve SUD [**1 mark**].

However, the CDI has a skewed focus towards social sustainability and does not cover all aspects of sustainable urban development [**1 mark**].

Numbers **lack meaningful context** [**1 mark**].

(e) Explain the impact of multiple deprivations, experienced by slum dwellers, on sustainable urban development. [5]

The slum dwellers occupy land illegally that are in zones with physical disadvantages compromising on personal safety [**1 mark**]. They are more susceptible to environmental risk which SDG target 11.1 [**1 additional mark**].

Many of these slums do not have access to clean and green spaces due to the location of the settlement [**1 mark**]

The lack of basic services in slums undermines public health, contributing toward higher morbidity and mortality as diseases are easily spread [**1 mark**]

The lack of such basic utilities such as a reliable energy source can limit economic opportunities [**1 mark**]

(f) Explain the role of human activities and associated feedback mechanisms in influencing global warming. [5]

Due to the presence of more greenhouse gas in the atmosphere, it leads to warming [**1 mark**]. Due to warming, ice melts to reveal open water, which has a lower albedo than ice. More solar energy is absorbed by the open water leading to further warmer [**1 additional mark**]

In addition, further warming of earth can result in more permafrost melting, releasing stored methane and carbon dioxide (greenhouse gases) into the atmosphere, and so producing more warming [**1 mark**]

Due to higher temperature, more water can be held in the atmosphere, leading to an increase in cloudiness increasing the atmospheric albedo [**1 mark**] which results in more reflection of insolation. This could thus lead to cooling with less heat absorbed [**1 additional mark**].

3. 'Urban reimaging strategies are successful in all places.' Evaluate this statement.

Candidates could approach the question by making a judgement on whether some strategies are more effective than others through a consideration of the relative strengths and limitations of the strategies. The strategies could be compared in relation to criteria such as costs, benefits and ability to target a wide range of issues. These strategies could be aimed at addressing different issues related to elderly such as those related to social, economic and environmental sustainability.

4. 'Partnership between firms and states is important to respond to climate change successfully.' Evaluate this statement.

Candidates could approach the question by making a judgement on whether partnership is important through a consideration of the scale of actions/outcomes, immediacy of the actors' actions/outcomes and how long lasting the actions/outcomes are. These actions could be aimed at addressing different responses to climate change such as those related to mitigation and adaptation to impacts of climate change.